

CSE30301: Basic Math for AI - Final Exam Solutions

Question 1 [30 points]

Question	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
Answer	T	F	F	T	F	F	T	F	F	T

- 1.1** True. The gradient points in the direction of steepest local increase.
- 1.2** False. The negative gradient gives the steepest local decrease, but it is not necessarily the fastest global path to the optimum.
- 1.3** False. A zero-gradient point can be a local minimum, local maximum, or saddle point.
- 1.4** True. For logistic regression with cross entropy loss, the gradient can be written as an error term multiplied by the input.
- 1.5** False. A larger learning rate can cause oscillation or divergence.
- 1.6** False. A small gradient norm can occur near a saddle point, plateau, or saturated region.
- 1.7** True. Maximum likelihood estimation is equivalent to minimizing the forward KL divergence from the empirical data distribution to the model distribution, up to a constant independent of the model parameters.
- 1.8** False. KL divergence is not symmetric in general.
- 1.9** False. Reverse KL is generally associated with mode-seeking behavior, not mode-covering.
- 1.10** True. Wasserstein distance can remain meaningful even when the supports of two distributions are disjoint.

Question 2 [25 points]

We have

$$p_i = \sigma(x_i^T \beta), \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)).$$

Thus,

$$\nabla_{\beta} p_i = p_i(1 - p_i)x_i.$$

2.1

The MSE loss is

$$J_{\text{MSE}}(\beta) = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2.$$

Therefore,

$$\nabla_{\beta} J_{\text{MSE}}(\beta) = \frac{1}{N} \sum_{i=1}^N 2(p_i - y_i) \nabla_{\beta} p_i.$$

Using $\nabla_{\beta} p_i = p_i(1 - p_i)x_i$,

$$\nabla_{\beta} J_{\text{MSE}}(\beta) = \frac{2}{N} \sum_{i=1}^N (p_i - y_i) p_i(1 - p_i)x_i.$$

2.2

The cross entropy loss is

$$J_{\text{CE}}(\beta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)].$$

For a single sample,

$$\frac{\partial}{\partial p_i} [-y_i \log p_i - (1 - y_i) \log(1 - p_i)] = -\frac{y_i}{p_i} + \frac{1 - y_i}{1 - p_i}.$$

Therefore,

$$\nabla_{\beta} J_{\text{CE}}(\beta) = \frac{1}{N} \sum_{i=1}^N \left(-\frac{y_i}{p_i} + \frac{1 - y_i}{1 - p_i} \right) p_i(1 - p_i)x_i.$$

Simplifying,

$$\left(-\frac{y_i}{p_i} + \frac{1 - y_i}{1 - p_i} \right) p_i(1 - p_i) = -y_i(1 - p_i) + (1 - y_i)p_i = p_i - y_i.$$

Thus,

$$\nabla_{\beta} J_{\text{CE}}(\beta) = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)x_i.$$

2.3

For MSE,

$$\nabla_{\beta} J_{\text{MSE}}(\beta) = \frac{2}{N} \sum_{i=1}^N (p_i - y_i) p_i(1 - p_i)x_i.$$

This gradient contains the extra factor

$$p_i(1 - p_i).$$

When the sigmoid is saturated, $p_i \approx 0$ or $p_i \approx 1$, and therefore

$$p_i(1 - p_i) \approx 0.$$

Hence, even if the prediction is wrong, the MSE gradient can become very small.

For cross entropy,

$$\nabla_{\beta} J_{\text{CE}}(\beta) = \frac{1}{N} \sum_{i=1}^N (p_i - y_i) x_i.$$

The sigmoid derivative factor is canceled. Therefore, when the prediction is wrong, $p_i - y_i$ can remain large, and cross entropy can still produce a useful gradient.

Question 3 [20 points]

The loss is

$$L(\theta_1, \theta_2) = \frac{1}{2}\theta_1^2 + \frac{1}{2}\theta_2^2.$$

3.1

The gradient is

$$\nabla L(\theta) = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

At

$$\theta = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

we get

$$\nabla L(\theta) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

3.2

Let

$$u = \begin{bmatrix} \cos \phi \\ \sin \phi \end{bmatrix}.$$

Then

$$D_u L(\theta) = \nabla L(\theta)^T u = \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} \cos \phi \\ \sin \phi \end{bmatrix}.$$

Thus,

$$D_u L(\theta) = 3 \cos \phi + 4 \sin \phi.$$

3.3

The fastest local increase direction is

$$\frac{\nabla L(\theta)}{\|\nabla L(\theta)\|}.$$

Therefore, the fastest local decrease direction is

$$u^* = -\frac{\nabla L(\theta)}{\|\nabla L(\theta)\|}.$$

Since

$$\nabla L(\theta) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \|\nabla L(\theta)\| = \sqrt{3^2 + 4^2} = 5,$$

we obtain

$$u^* = -\frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}.$$

3.4

For a small movement $\Delta = \epsilon u$,

$$L(\theta + \epsilon u) \approx L(\theta) + \epsilon \nabla L(\theta)^T u.$$

To decrease the loss most rapidly, we choose the unit vector u that minimizes

$$\nabla L(\theta)^T u.$$

This is achieved by

$$u = -\frac{\nabla L(\theta)}{\|\nabla L(\theta)\|}.$$

Therefore, gradient descent updates parameters in the negative gradient direction:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t).$$

Question 4 [20 points]

4.1

The objective is

$$L(\theta) = a\theta^2.$$

Since $a > 0$, the minimizer satisfies

$$\frac{dL}{d\theta} = 2a\theta = 0.$$

Thus,

$$\theta^* = 0.$$

The gradient descent update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta \cdot 2a\theta^{(t)}.$$

Therefore,

$$\theta^{(t+1)} = (1 - 2a\eta)\theta^{(t)}.$$

Since $\theta^* = 0$,

$$\theta^{(t+1)} - \theta^* = (1 - 2a\eta)(\theta^{(t)} - \theta^*).$$

Hence,

$$c = 1 - 2a\eta.$$

4.2

From 4.1,

$$\theta^{(t+1)} - \theta^* = c(\theta^{(t)} - \theta^*).$$

Applying this recursively gives

$$\begin{aligned}\theta^{(1)} - \theta^* &= c(\theta^{(0)} - \theta^*), \\ \theta^{(2)} - \theta^* &= c(\theta^{(1)} - \theta^*) = c^2(\theta^{(0)} - \theta^*).\end{aligned}$$

In general,

$$\theta^{(t)} - \theta^* = c^t(\theta^{(0)} - \theta^*).$$

4.3

For $\theta^{(t)}$ to converge to θ^* , we need

$$\theta^{(t)} - \theta^* = c^t(\theta^{(0)} - \theta^*) \rightarrow 0.$$

This holds if

$$|c| < 1.$$

Since

$$c = 1 - 2a\eta,$$

the convergence condition is

$$|1 - 2a\eta| < 1.$$

Equivalently,

$$-1 < 1 - 2a\eta < 1.$$

Subtracting 1,

$$-2 < -2a\eta < 0.$$

Dividing by $-2a$ reverses the inequality signs because $a > 0$:

$$0 < \eta < \frac{1}{a}.$$

Therefore, gradient descent converges to the minimizer if

$$0 < \eta < \frac{1}{a}.$$

More specifically:

- If $0 < \eta < \frac{1}{2a}$, then $0 < c < 1$, so the update converges monotonically.
- If $\eta = \frac{1}{2a}$, then $c = 0$, so the update reaches the minimizer in one step.
- If $\frac{1}{2a} < \eta < \frac{1}{a}$, then $-1 < c < 0$, so the update converges with oscillation.
- If $\eta > \frac{1}{a}$, then $|c| > 1$, so the update diverges.

4.4

For

$$L(\theta_1, \theta_2) = 10\theta_1^2 + \theta_2^2,$$

each coordinate behaves like a one-dimensional quadratic.

For the first coordinate,

$$L_1(\theta_1) = 10\theta_1^2,$$

so $a = 10$. The convergence condition is

$$0 < \eta < \frac{1}{10}.$$

For the second coordinate,

$$L_2(\theta_2) = \theta_2^2,$$

so $a = 1$. The convergence condition is

$$0 < \eta < 1.$$

Thus, the first coordinate requires a much smaller learning rate than the second coordinate. If η is chosen large enough for fast progress along the low-curvature direction, it may be too large for the high-curvature direction and cause oscillation or divergence. If η is chosen small enough to be stable along the high-curvature direction, progress along the low-curvature direction may become very slow.

Question 5 [30 points]

The objective is

$$L(\boldsymbol{\theta}) = 10\theta_1^2 + \theta_2^2.$$

Therefore,

$$\nabla L(\boldsymbol{\theta}) = \begin{bmatrix} 20\theta_1 \\ 2\theta_2 \end{bmatrix}.$$

5.1

At

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

we have

$$\nabla L(\boldsymbol{\theta}^{(0)}) = \begin{bmatrix} 20 \\ 2 \end{bmatrix}.$$

Using vanilla gradient descent with $\eta = 0.1$,

$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - 0.1 \begin{bmatrix} 20 \\ 2 \end{bmatrix}.$$

Thus,

$$\boldsymbol{\theta}^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0.2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0.8 \end{bmatrix}.$$

5.2

At

$$\boldsymbol{\theta}^{(1)} = \begin{bmatrix} -1 \\ 0.8 \end{bmatrix},$$

the gradient is

$$\nabla L(\boldsymbol{\theta}^{(1)}) = \begin{bmatrix} -20 \\ 1.6 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{\theta}^{(2)} = \begin{bmatrix} -1 \\ 0.8 \end{bmatrix} - 0.1 \begin{bmatrix} -20 \\ 1.6 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.64 \end{bmatrix}.$$

At

$$\boldsymbol{\theta}^{(2)} = \begin{bmatrix} 1 \\ 0.64 \end{bmatrix},$$

the gradient is

$$\nabla L(\boldsymbol{\theta}^{(2)}) = \begin{bmatrix} 20 \\ 1.28 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{\theta}^{(3)} = \begin{bmatrix} 1 \\ 0.64 \end{bmatrix} - 0.1 \begin{bmatrix} 20 \\ 1.28 \end{bmatrix} = \begin{bmatrix} -1 \\ 0.512 \end{bmatrix}.$$

Thus, the first coordinate oscillates:

$$1 \rightarrow -1 \rightarrow 1 \rightarrow -1.$$

5.3

The main problem is that the same learning rate is used for coordinates with very different gradient scales.

The Hessian is

$$\nabla^2 L = \begin{bmatrix} 20 & 0 \\ 0 & 2 \end{bmatrix}.$$

The curvature in the θ_1 direction is much larger than that in the θ_2 direction. With $\eta = 0.1$,

$$\theta_1^{(t+1)} = \theta_1^{(t)} - 0.1 \cdot 20\theta_1^{(t)} = -\theta_1^{(t)}.$$

Thus, the first coordinate oscillates.

For the second coordinate,

$$\theta_2^{(t+1)} = \theta_2^{(t)} - 0.1 \cdot 2\theta_2^{(t)} = 0.8\theta_2^{(t)}.$$

Thus, the second coordinate decreases smoothly but more slowly. This is a coordinate-wise gradient-scale or curvature imbalance problem.

5.4

We want to keep the same learning rate $\eta = 0.1$ and design a coordinate-wise scaled update such that

$$\theta_1^{(t+1)} = 0.5\theta_1^{(t)}, \quad \theta_2^{(t+1)} = 0.5\theta_2^{(t)}.$$

Consider an update of the form

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta D \nabla L(\boldsymbol{\theta}^{(t)}),$$

where

$$D = \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix}.$$

Since

$$\nabla L(\boldsymbol{\theta}^{(t)}) = \begin{bmatrix} 20\theta_1^{(t)} \\ 2\theta_2^{(t)} \end{bmatrix},$$

the update becomes

$$\theta_1^{(t+1)} = \theta_1^{(t)} - 0.1d_1 \cdot 20\theta_1^{(t)} = (1 - 2d_1)\theta_1^{(t)}.$$

We want

$$1 - 2d_1 = 0.5,$$

so

$$d_1 = 0.25.$$

Similarly,

$$\theta_2^{(t+1)} = \theta_2^{(t)} - 0.1d_2 \cdot 2\theta_2^{(t)} = (1 - 0.2d_2)\theta_2^{(t)}.$$

We want

$$1 - 0.2d_2 = 0.5,$$

so

$$d_2 = 2.5.$$

Therefore, one valid modified update is

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - 0.1 \begin{bmatrix} 0.25 & 0 \\ 0 & 2.5 \end{bmatrix} \nabla L(\boldsymbol{\theta}^{(t)}).$$

Equivalently,

$$D = \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{5}{2} \end{bmatrix}.$$

This update rescales the coordinates so that both coordinates shrink by the same factor 0.5 at each iteration. It reduces oscillation in the high-curvature direction while avoiding unnecessarily slow progress in the low-curvature direction.

5.5

From 5.4, the modified update gives

$$\boldsymbol{\theta}^{(t+1)} = 0.5\boldsymbol{\theta}^{(t)}.$$

Starting from

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

we obtain

$$\boldsymbol{\theta}^{(1)} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix},$$

$$\boldsymbol{\theta}^{(2)} = \begin{bmatrix} 0.25 \\ 0.25 \end{bmatrix},$$

and

$$\boldsymbol{\theta}^{(3)} = \begin{bmatrix} 0.125 \\ 0.125 \end{bmatrix}.$$

Question 6 [20 points]

Let

$$q(x) = \mathcal{N}(x; \mu, \sigma^2).$$

Then

$$q(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Taking logarithm,

$$\log q(x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}.$$

For any density $p(x)$ satisfying

$$\mathbb{E}_p[X] = \mu, \quad \text{Var}_p(X) = \sigma^2,$$

we have

$$\mathbb{E}_p[(X-\mu)^2] = \sigma^2.$$

Therefore,

$$\mathbb{E}_p[\log q(X)] = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \mathbb{E}_p[(X-\mu)^2].$$

Thus,

$$\mathbb{E}_p[\log q(X)] = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2}.$$

Using the non-negativity of KL divergence,

$$D_{KL}(p||q) \geq 0.$$

Now,

$$D_{KL}(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx.$$

This can be written as

$$D_{KL}(p||q) = \int p(x) \log p(x) dx - \int p(x) \log q(x) dx.$$

Since

$$H(p) = -\int p(x) \log p(x) dx,$$

we have

$$D_{KL}(p||q) = -H(p) - \mathbb{E}_p[\log q(X)].$$

Since $D_{KL}(p||q) \geq 0$,

$$-H(p) - \mathbb{E}_p[\log q(X)] \geq 0.$$

Therefore,

$$H(p) \leq -\mathbb{E}_p[\log q(X)].$$

Using the value computed above,

$$H(p) \leq \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2}.$$

Since

$$\frac{1}{2} = \frac{1}{2} \log e,$$

we get

$$H(p) \leq \frac{1}{2} \log(2\pi e \sigma^2).$$

For $p = q = \mathcal{N}(\mu, \sigma^2)$, equality holds because

$$D_{KL}(q||q) = 0.$$

Therefore, the Gaussian distribution

$$p^*(x) = \mathcal{N}(x; \mu, \sigma^2)$$

achieves the maximum entropy.

The corresponding maximum entropy value is

$$H(p^*) = \frac{1}{2} \log(2\pi e \sigma^2).$$

Question 7 [30 points]

The target distribution is

$$p(x) = \frac{1}{4}\mathcal{N}(x; -2, 1) + \frac{3}{4}\mathcal{N}(x; 2, 1).$$

The model distribution is

$$q_{\mu,\sigma}(x) = \mathcal{N}(x; \mu, \sigma^2).$$

7.1

We minimize

$$D_{KL}(p||q_{\mu,\sigma}) = \mathbb{E}_p[\log p(X)] - \mathbb{E}_p[\log q_{\mu,\sigma}(X)].$$

The first term does not depend on μ or σ . Therefore, minimizing forward KL is equivalent to maximizing

$$\mathbb{E}_p[\log q_{\mu,\sigma}(X)].$$

Since

$$\log q_{\mu,\sigma}(x) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2},$$

we need to maximize

$$-\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \mathbb{E}_p[(X - \mu)^2].$$

For fixed σ^2 , this is equivalent to minimizing

$$\mathbb{E}_p[(X - \mu)^2].$$

Differentiating,

$$\frac{d}{d\mu} \mathbb{E}_p[(X - \mu)^2] = \mathbb{E}_p[-2(X - \mu)] = -2(\mathbb{E}_p[X] - \mu).$$

Setting this equal to zero gives

$$\mu^* = \mathbb{E}_p[X].$$

7.2

From 7.1, the optimal mean is

$$\mu^* = \mathbb{E}_p[X].$$

Now optimize with respect to σ^2 . Let

$$s = \sigma^2.$$

The relevant objective is

$$-\frac{1}{2} \log(2\pi s) - \frac{1}{2s} \mathbb{E}_p[(X - \mu^*)^2].$$

Let

$$A = \mathbb{E}_p[(X - \mu^*)^2].$$

Then the objective becomes

$$-\frac{1}{2} \log(2\pi s) - \frac{A}{2s}.$$

Differentiate with respect to s :

$$-\frac{1}{2s} + \frac{A}{2s^2}.$$

Setting this equal to zero,

$$-\frac{1}{2s} + \frac{A}{2s^2} = 0.$$

Multiplying by $2s^2$,

$$-s + A = 0.$$

Thus,

$$s = A.$$

Therefore,

$$(\sigma^*)^2 = \mathbb{E}_p[(X - \mu^*)^2] = \text{Var}_p(X).$$

7.3

Let $Z \in \{1, 2\}$ denote the mixture component:

$$P(Z = 1) = \frac{1}{4}, \quad P(Z = 2) = \frac{3}{4}.$$

Then

$$X|Z = 1 \sim \mathcal{N}(-2, 1), \quad X|Z = 2 \sim \mathcal{N}(2, 1).$$

Thus,

$$\mathbb{E}[X|Z = 1] = -2, \quad \mathbb{E}[X|Z = 2] = 2.$$

Therefore,

$$\mu^* = \mathbb{E}[X] = \frac{1}{4}(-2) + \frac{3}{4}(2) = -\frac{1}{2} + \frac{3}{2} = 1.$$

7.4

Using the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X|Z)] + \text{Var}(\mathbb{E}[X|Z]).$$

First,

$$\text{Var}(X|Z = 1) = 1, \quad \text{Var}(X|Z = 2) = 1.$$

Therefore,

$$\mathbb{E}[\text{Var}(X|Z)] = \frac{1}{4} \cdot 1 + \frac{3}{4} \cdot 1 = 1.$$

Next,

$$\mathbb{E}[X|Z = 1] = -2, \quad \mathbb{E}[X|Z = 2] = 2,$$

and

$$\mathbb{E}[X] = 1.$$

Thus,

$$\text{Var}(\mathbb{E}[X|Z]) = \frac{1}{4}(-2 - 1)^2 + \frac{3}{4}(2 - 1)^2.$$

So,

$$\text{Var}(\mathbb{E}[X|Z]) = \frac{1}{4}(9) + \frac{3}{4}(1) = \frac{9}{4} + \frac{3}{4} = 3.$$

Therefore,

$$\text{Var}(X) = 1 + 3 = 4.$$

Hence,

$$(\sigma^*)^2 = 4.$$

The final forward KL optimal model is

$$q^*(x) = \mathcal{N}(x; 1, 4).$$

Question 8 [25 points]

Let

$$P(x) = \text{Unif}[0, 1], \quad Q_\mu(x) = \text{Unif}[\mu, \mu + 1].$$

8.1

For $\mu = \frac{1}{2}$,

$$Q_\mu(x) = \text{Unif}\left[\frac{1}{2}, \frac{3}{2}\right].$$

The supports are

$$\text{supp}(P) = [0, 1], \quad \text{supp}(Q_\mu) = \left[\frac{1}{2}, \frac{3}{2}\right].$$

The overlap region is

$$\left[\frac{1}{2}, 1\right],$$

and the non-overlap regions are

$$\left[0, \frac{1}{2}\right) \quad \text{and} \quad \left(1, \frac{3}{2}\right].$$

First, consider forward KL:

$$D_{KL}(P||Q_\mu) = \int P(x) \log \frac{P(x)}{Q_\mu(x)} dx.$$

On the region

$$x \in \left[0, \frac{1}{2}\right),$$

we have

$$P(x) = 1, \quad Q_\mu(x) = 0.$$

Therefore,

$$P(x) \log \frac{P(x)}{Q_\mu(x)} = \log \frac{1}{0} = \infty.$$

Hence,

$$D_{KL}(P||Q_\mu) = \infty.$$

Now compute JSD. Let

$$M = \frac{1}{2}(P + Q_\mu).$$

On the overlap region

$$x \in \left[\frac{1}{2}, 1\right],$$

we have

$$P(x) = 1, \quad Q_\mu(x) = 1, \quad M(x) = 1.$$

Thus the KL contribution from the overlap region is zero.

On the P -only region

$$x \in \left[0, \frac{1}{2}\right),$$

we have

$$P(x) = 1, \quad Q_\mu(x) = 0, \quad M(x) = \frac{1}{2}.$$

Thus,

$$\log \frac{P(x)}{M(x)} = \log \frac{1}{1/2} = \log 2.$$

The length of this region is $\frac{1}{2}$, so

$$D_{KL}(P||M) = \frac{1}{2} \log 2.$$

On the Q_μ -only region

$$x \in \left(1, \frac{3}{2}\right],$$

we have

$$Q_\mu(x) = 1, \quad P(x) = 0, \quad M(x) = \frac{1}{2}.$$

Similarly,

$$D_{KL}(Q_\mu||M) = \frac{1}{2} \log 2.$$

Therefore,

$$JSD(P, Q_\mu) = \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(Q_\mu||M).$$

Substituting,

$$JSD(P, Q_\mu) = \frac{1}{2} \cdot \frac{1}{2} \log 2 + \frac{1}{2} \cdot \frac{1}{2} \log 2 = \frac{1}{2} \log 2.$$

8.2

For $\mu = \frac{1}{2}$, forward KL is not appropriate because

$$D_{KL}(P||Q_\mu) = \infty.$$

This happens because there is a region where $P(x) > 0$ but $Q_\mu(x) = 0$.

By contrast, JSD is finite:

$$JSD(P, Q_\mu) = \frac{1}{2} \log 2.$$

Therefore, in this partial-overlap case, JSD is more appropriate than KL because it avoids the infinite value caused by support mismatch.

8.3

For $\mu = 2$,

$$Q_\mu(x) = \text{Unif}[2, 3].$$

The supports are fully separated:

$$[0, 1] \cap [2, 3] = \emptyset.$$

Forward KL is still inappropriate because for every

$$x \in [0, 1],$$

we have

$$P(x) = 1, \quad Q_\mu(x) = 0.$$

Thus,

$$D_{KL}(P||Q_\mu) = \infty.$$

Therefore, KL does not provide a finite objective.

For JSD, the value is finite. Since the supports are fully separated, the mixture distribution is

$$M(x) = \frac{1}{2}$$

on each support. Thus,

$$D_{KL}(P||M) = \log 2, \quad D_{KL}(Q_\mu||M) = \log 2.$$

Therefore,

$$JSD(P, Q_\mu) = \frac{1}{2} \log 2 + \frac{1}{2} \log 2 = \log 2.$$

However, for any $\mu \geq 1$, the two supports remain disjoint. Hence,

$$JSD(P, Q_\mu) = \log 2$$

is constant over that region. Therefore,

$$\frac{d}{d\mu} JSD(P, Q_\mu) = 0$$

for fully separated supports.

Thus, neither KL nor JSD is appropriate in the fully separated case:

- KL is infinite and not useful as a finite optimization objective.
- JSD is finite but saturated, giving no useful gradient direction for moving Q_μ toward P .